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Martinize2 and Vermouth: Unified Framework for Topology Generation

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Abstract

Ongoing advances in force field and computer hardware development enable the use of molecular dynamics (MD) to simulate increasingly complex systems with the ultimate goal of reaching cellular complexity. At the same time, rational design by high-throughput (HT) simulations is another forefront of MD. In these areas, the Martini coarse-grained force field, especially the latest version (*i.e.* v3), is being actively explored because it offers enhanced spatial-temporal resolution. However, the automation tools for preparing simulations with the Martini force field, accompanying the previous version, were not designed for HT simulations or studies of complex cellular systems. Therefore, they become a major limiting factor. To address these shortcomings, we present the open-source *vermouth* python library. *Vermouth* is designed to become the unified framework for developing programs, which prepare, run, and analyze Martini simulations of complex systems. To demonstrate the power of the *vermouth* library, the *martinize2* program is showcased as a generalization of the *martinize* script, originally aimed to set up simulations of proteins. In contrast to the previous version, *martinize2* automatically handles protonation states in proteins and post-translation modifications, offers more options to fine-tune structural biases such as the elastic network, and can convert nonprotein molecules such as ligands. Finally, *martinize2* is used in two high-complexity benchmarks. The entire I-TASSER protein template database as well as a subset of 200,000 structures from the AlphaFold Protein Structure Database are converted to CG resolution and we illustrate how the checks on input structure quality can safeguard HT applications.

eLife assessment

The authors present a **valuable** computational platform, which aims to automate the workflow for coarse-grained simulations of biomolecules in the framework of the popular MARTINI model. The capability of the platform has been **convincingly** demonstrated by the application to a large number of proteins as well as macrocycles and polymers. On the other hand, because the developments have largely been based on the MARTINI model, some might argue that the general impact on the multi-scale simulation community is limited, leaving the support for the claimed significance **incomplete**.

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Introduction

Molecular dynamics (MD) has grown to be a valuable and powerful tool in studying a variety of systems in molecular detail. Advances in force fields and computer hardware have enabled the use of MD in increasingly complex systems, exemplified by recent simulations of, *e.g.* realistic cell membranes[1,2], virus particles[2,3], and even complete aerosol droplets[4]. However, there is a growing interest in studying systems of even greater complexity, culminating in molecularly detailed simulations of whole organelles[5,6] and the set goal of simulating entire cells[7–9]. Moreover, the growing demand for computer-aided rational design relies on high-throughput (HT) simulations with millions of systems simulated in parallel.[10–12] Currently, the computational demand of MD methods representing all atoms explicitly severely limits the access to the spatial-temporal resolution needed to simulate the aforementioned systems. Coarse-grained (CG) MD methods overcome this challenge by grouping several atoms into one effective interaction site called bead and thus reducing the number of degrees of freedom that have to be simulated.

Among the most popular CG methods is the Martini force field.[13,14] Within the scope of the Martini force field about 2–5 non-hydrogen atoms are grouped into one bead. Nonbonded interactions between beads are defined in a discrete interaction table calibrated to reproduce thermodynamic data, whereas bonded interactions are matched to underlying atomistic reference simulations. Molecule parameters created following this approach are transferable between different systems and chemical contexts.[13,14] This transferability-based approach allows Martini simulations to easily reach the aforementioned complexity scale. However, to really prepare Martini for the HT and whole cell scale simulation era, automated workflows that enable fast and efficient setup of complex systems are of fundamental importance.

The Martini community has a long-standing history of easy-to-use and freely accessible scripts and programs, which helped researchers to set up, run, analyze, and backmap simulations. A non-exhaustive overview can be found in our recent review of the 20-year history of Martini.[15] However, the codes and scripts developed so far present a collection of separate scripts that share no common framework or backend even though they share many common operations such as resolution transformation or mapping of coordinates. In addition, input files which define molecule parameters or fragments thereof, are not transferable between the tools, with each one of them often defining their own input file formats. We consider that unifying the operations as well as input streams into a single framework will speed up program development and also the robustness of code design to bugs. In addition, it will allow the implementation of modern software techniques such as code review, continuous integration (CI) testing, and version control which generally improve code quality and resilience.[16]

Designing and coding a unified framework to support general Martini software development is a massive undertaking with many facets as the original scripts and programs deal with different stages of MD simulations. To start the development, we focused the design of the framework on topology generation. A topology lies at the heart of each simulation and defines the starting coordinates as well as input parameters for the simulation. For example, to run protein simulations within Martini, a script called *martinize*[17] takes atomistic protein coordinates, maps them to the coarse-grained resolution, and generates the protein molecule definitions from building blocks. This workflow is quite classic and underlies many scripts and programs for topology generation both at the coarse-grained but also at the all-atom level.[17–36] With the latest release of version 3 of Martini (M3), proteins have been thoroughly reparametrized.[13] The new capabilities of M3 proteins are exemplified by their use of HT drug binding assays[11,37], which are an essential step in computer-aided drug design (CADD). Part of the improved protein properties come from the redefined Martini interaction table. However, another part of the improvement is the result of protein-specific methods such as the use of structure-biased dihedrals[38] (often referred to as side-chain corrections), specific elastic networks[39], or integration of Go-like models.[40] All features are additional specific biasing steps applied after the generation of the original topology file for the protein and are not part of the capabilities of the previous *martinize* script. Hence, we choose to co-develop a unified framework for topology generation together with a new *martinize* version, *martinize2*.

In this paper, we present the VERsatile MOdular Universal Transformation Helper (*vermouth*) library, a general python framework aiding in the design of programs that can create topologies for complex systems at all-atom (AA), united-atom (UA), and CG resolution. On top of *vermouth*, we built the *martinize2* program, as the successor of the *martinize* script[17,34]. The goal of *martinize2* is to encompass all functionality required to generate Martini protein parameters (supporting the older versions Martini 2[17,39,41] as well as the latest Martini 3) and be compatible with high-throughput workflows as needed in CADD approaches based on Martini. In addition, both *vermouth* and *martinize2* are designed to have sufficient flexibility and robustness to ready Martini for the era of high-throughput high complexity simulations.

Finally, we note that much of the progress of Martini has resulted from an active community of researchers contributing scripts, programs, and parameters. However, as is the case for most research software in the field they often fail to adhere to the principles of FAIR: findability, accessibility, interoperability, and reusability.[42–44] The FAIR principles[43], originally designed to improve data management and reproducibility in science, have recently been extended to research software in a more general sense. This extension is aimed at fostering more sustainable software development in science.[42] To meet these standards the software tools we present here are distributed under the permissive open-source Apache 2.0 license on GitHub and are developed using contemporary software development practices, such as continuous integration testing. To make adoption as easy as possible, they have few dependencies, are distributed through the Python Package Index, and can be installed using the Python package manager *pip*. Other researchers are encouraged and welcome to contribute parameters and code as outlined in our contribution workflow.

Results

In this section, we first outline the design and API of the *vermouth* library. Then we discuss how the *vermouth* library is used to construct a pipeline for generating protein input parameters for the Martini force field. This pipeline constitutes the new *martinize2* program. Finally, we present some benchmarks and selected test cases to demonstrate the capabilities of *martinize2* and assess its fitness for generating complex system topologies and high-throughput workflows, surpassing the capabilities of the previous *martinize* script.

The *Vermouth* Library

In the literature, many scripts and programs have been described that can create topologies for linear molecules and some specific software exists that also handles branched molecules such as carbohydrates[24], or dendritic polymers[25]. However, to the best of our knowledge, there is at present not a general program that can create topologies from atomistic structures for any type of system, and at any resolution, presenting an extendable and stable API. Based on the existing software, we can, however, define a number of required and desirable features for such a general program and library to have: 1) it must be force field and resolution agnostic; 2) it must be MD engine agnostic; 3) it must use data files that can be checked, made and modified by users, and 4) it must be able to process any type of molecule or polymer, be it linear, cyclic, branched, or dendrimeric, and mixtures thereof.

To start designing a library that can fulfill the above requirements we note that most workflows used for topology generation can be decomposed into six fundamental stages (**Figure 1**): First, reading input data, usually an atomistic coordinate file (e.g. from the protein data bank); second, identifying the parsed atoms, to find how they correspond to the atoms in the data files describing the building blocks; third, optionally a resolution transformation step; fourth, the generation of the actual topology and assigning particle types and bonded interactions; fifth, any type of post-processing; and finally, sixth, writing the required output files. Even though these stages are generally shared for topology generation pipelines, they also apply to other workflows commonly encountered in Martini programs. Especially, stages 1, 3, 5, and 6 can be found in almost all Martini programs, which generate simulation input files in the broader sense.[17,45–47] Separating these stages, therefore, helps to define an API with data structures and independent processes, which optimally support such workflows. In addition, the clear distinction in stages helps to externalize any data files, which can be edited by the user or force field developers. *Vermouth* is built on the idea and definition of processors, which are tasks arranged in a pipeline. This design was inspired by the ubiquitous workflow managers available in the field.[48] We formalize the idea of processors by introducing an abstract base class the ***Processor***. New pipeline stages can be created as subclasses of this base class. All ***Processors*** operate on the central data structure class ***System***, which contains any number of ***Molecule*** data structures (see **Figure 2**). A ***Molecule*** is defined as the graph of a molecule or assembly of molecules, which are connected by bonded interactions. The nodes of a ***Molecule*** usually correspond to atoms or coarse-grain beads but can be any form of particle as defined by the force field.

Nodes can have attributes that describe additional information such as a residue name or charge. However, only the atom name, residue name, and residue number are required as attributes. In addition, the edges of the ***Molecule*** follow the connectivity as defined by bonds, angles, or other bonded interactions. For example, two protein chains connected by a disulfide bridge would be considered a single ***Molecule***. In contrast, a cofactor, which is only interacting via non-bonded interactions, would be a separate ***Molecule***. Operations on ***Molecules*** usually add or remove bonded interactions or change node attributes. For convenience, ***Processors*** can also operate on a collection of molecules, which are defined by the ***System*** class (see **Figure 2**). A list of all available processors is given in the documentation.

Processors operate on ***Molecules***. However, often additional data is required to perform the pipeline as defined by the ***Processor***. The additional data can be provided in the form of one of the four other main data structures (***Blocks, Links, Modifications, Mappings***) or arguments of the processors that can be set in a script or via the command line interface. These four other data structures contain all molecular level information required to fully define and/or modify a topology for any type of MD code (e.g., atom types, bonded interactions, and positions) as well as enable transformations between topologies. For example, a ***Mapping*** consists of two molecular fragments at different resolutions and a correspondence between their particles. In contrast,

Figure 1

Fundamental stages in topology generation from atomistic structures.

First, the provided input is parsed (step 1). Second, for every parsed residue its atoms are identified and, if needed, atom names are corrected and missing atoms are added (step 2). Third, mappings are taken from the library and a resolution transformation to the required output resolution is performed (step 3). Fourth, intra-residue interactions are added from blocks taken from the library, and inter-residue interactions are added from links taken from the library (step 4). Fifth, optionally, post-processing is performed to add e.g. *an elastic network (step 5). Finally, the produced topology is written to output files (step 6).*

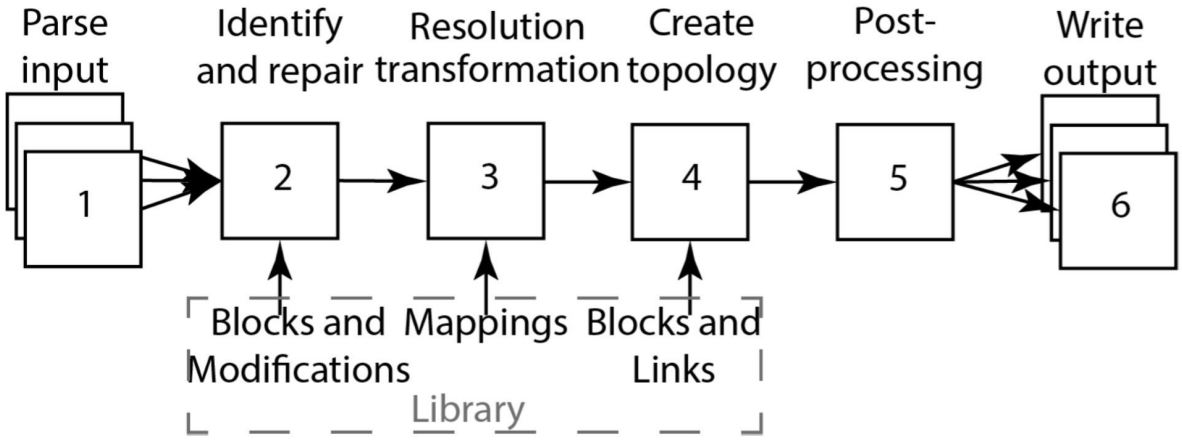
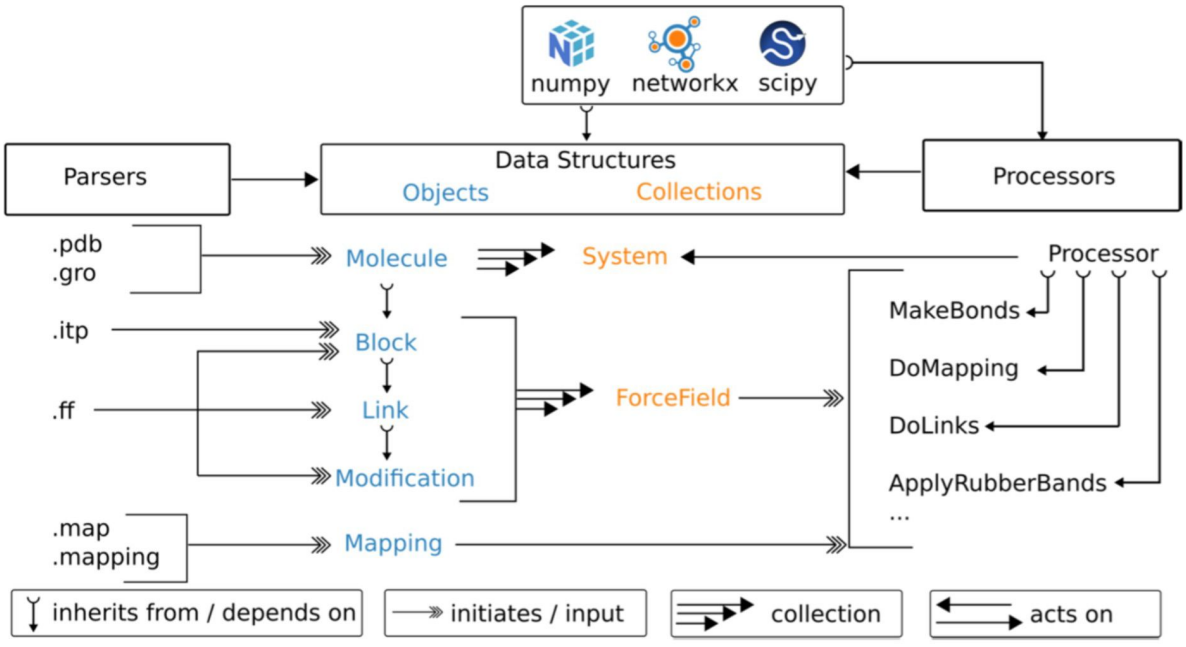


Figure 2

Organization of the vermouth library.

The vermouth library defines 5 types of data structures (blue) to store molecular information and force field information. For convenience, it also defines two collection classes (orange) composed of several data-structure instances. Data structures are initiated or get input from parsers, which read 6 types of data files (see ***Table S1*** *for more details on file types). The central data structure(s) are* ***Molecule*** and ***System.*** These *are changed, updated, or transformed by so-called* ***Processor*** *classes, which take force field data as input. Parsers, data structures, and* ***Processors*** *only depend on three libraries as shown. At the moment vermouth also exposes four types of writers (not shown here) to go along with the parsers (see* ***Table S2****).*



Blocks, Links, and Modifications are graphs, which describe these molecular fragments, the links between those, and possible changes to fragments respectively. They are all subclasses of a ***Molecule*** and an extension of the graph class from the networkx library[49] (see **Figure 2**).

As shown in **Figure 2**, to make the data structures that are force field specific (***Blocks, Links, Modifications***) easier to use, *vermouth* offers a second collection class called a ***ForceField***. Every molecule must have a ***ForceField*** associated with it. Additional information on the data structures is given in the documentation.

Finally, the *vermouth* library also contains a number of parsers that return instances of the data structures from common input file formats. For example, the in-house ff format defines ***Blocks***, ***Links***, and ***Modifications***, while the backwards style mapping format can be read to return an instance of the ***Mapping*** class. **Table S1** in the Supporting Information summarizes all input parsers as well as the format and data structure they return. We note that *vermouth* is also able to read content associated with the '[molecules]' directive of the GROMACS topology file, which is colloquially referred to as itp file. This allows to directly manipulate GROMACS molecule files within *vermouth*. We note that as neither parsing nor the ***Molecule*** itself depends on GROMACS code, the library can easily be extended to other MD engines.

Martinize2

Martinize2 is a pipeline constructed of vermouth ***Processors*** with a command line interface (CLI), with the purpose of transforming atomistic structure data to a CG Martini topology including both coordinates and simulation parameters. *Martinize2* is the successor of the *martinize* script, which was used for generating input parameters for Martini version 2 proteins, DNA, or RNA. However, different branches had to be used for proteins and DNA (martinize.py[17], martinize-dna.py[34]) or RNA[36]. In contrast, *martinize2* is designed to generate topologies for the Martini force field for proteins, DNA, and in principle any other arbitrarily complex molecule.

Martinize2 consists of different ***Processors*** which fulfill the basic stages of topology generation as shown in **Figure 1**. We note that the design of *martinize2* is general and applies to arbitrarily complex polymers consisting of arbitrary monomeric repeat units (MRUs). However, to increase the readability of the following sections the layout of the program is described in terms of residues in proteins.

The *martinize2* pipeline starts by reading an atomistic structure, which describes a single molecule (*e.g.*, protein) or assembly of any size. Subsequently, bonds between the atoms are inferred either by distance calculation, atom names within residues, or using CONECT records of the PDB file. All atoms that are connected by bonds form a ***Molecule***. Thus, *martinize2* creates a ***System*** of ***Molecules*** at the atomistic resolution at the end of the input reading stage. In stage 2, *Identify and Repair*, each residue of each molecule is compared against its canonical definition. Canonical definitions are selected by residue name from the library files. This comparison identifies missing or additional atoms on a residue and fixes all atom names (**Figure 3a**). To efficiently do these comparison operations, *martinize2* relies on a number of algorithms coming from graph theory (*e.g.* subgraph isomorphism), which reduces the dependence on accurate atom names, since these occasionally differ based on the source of the input structure. Which algorithms are used in the code is described in more detail in the Supporting Information. Once the residues have their canonical atom names, *martinize2* checks if the missing or additional atoms are described by any of the modification files (**Figure 3b**). Modifications describe changes in residues from their canonical form, *e.g.* different protonation states, termini, or post-translational modifications (PTMs), and the effect these have on the topology.

After completing the repair stage everything is in place to perform the mapping to coarsegrained resolution. The mapping descriptions are read from the mapping input files in the library and tie together residue definitions at the all-atom and CG level and the correspondence between them

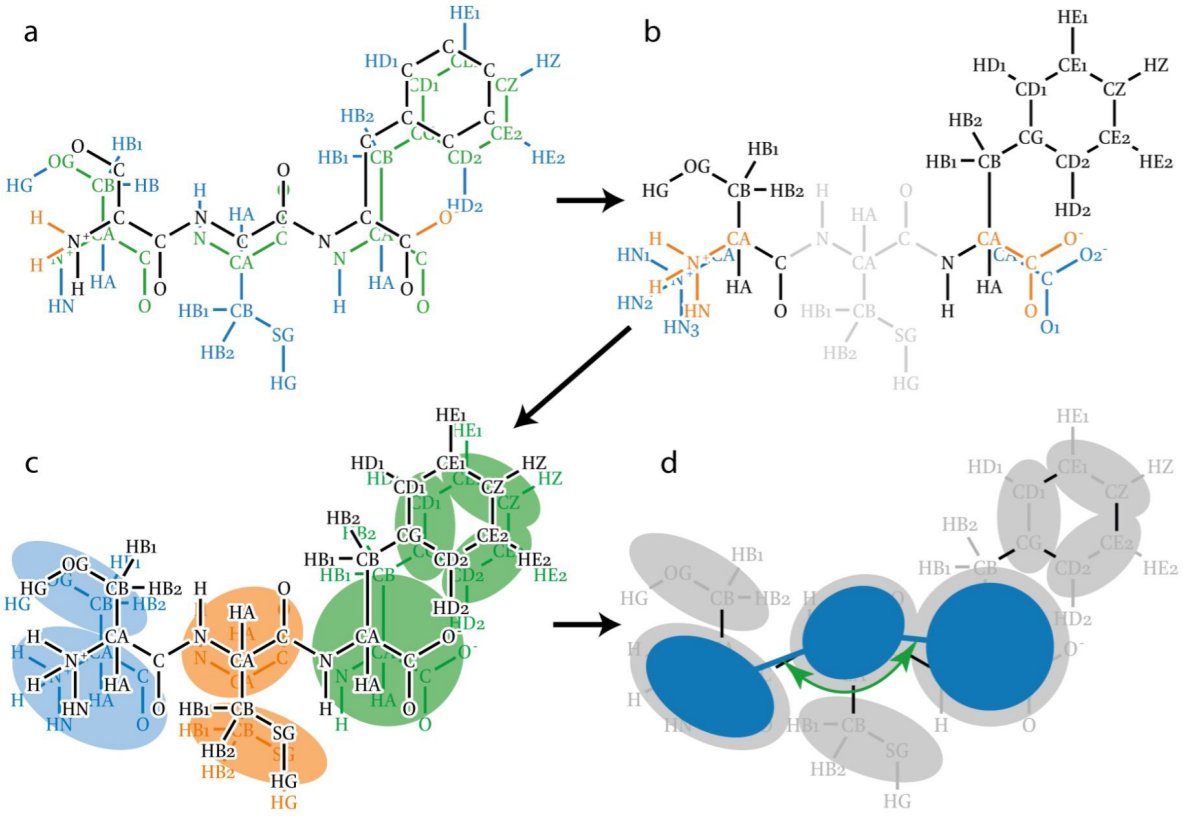


Figure 3

Illustration of atom recognition, mapping, and linking in topology generation.

a) To identify all atoms in the input molecule (black and orange) every residue in the molecule is overlaid with its canonical reference (blue and green). Atoms are recognized when they overlap with atoms in the reference (green). Atoms not present in the molecule are also identified (blue) and will be added later. Finally, atoms in the molecule not described by the canonical references are also labeled (orange) so that they may be identified later. b) Identifying the terminal atoms that are not part of the canonical residues. The modification templates are depicted in blue and the atoms they match in orange. The cysteine does not participate since it does not carry any unexpected atoms, and is depicted in grey for clarity. c) Mappings (blue, red, and green) describe a molecular fragment at two different resolutions and a correspondence between their particles. The correspondence is depicted approximately here. The mappings are applied to the molecule (black). d) Example of applying a Link. The link depicted (dark blue) adds an angle potential over CG backbone beads.

(**Figure 3c**). Mapping to CG level in *martinize2* is done with a multistep subgraph isomorphism procedure, which is general enough to cover edge cases such as when mappings span multiple atomistic residues. A detailed description is provided in the Supporting Information. The mapping provides a ***System*** of ***Molecules*** at the CG level. These molecules already define all bonded interactions within the residues as well as the coordinates of the CG system. To generate the interactions that link the residues, a simple graph matching with library link definitions is done in the create-topology stage (**Figure 3d**). Finally, after that, we end up with the full CG topology, which is ready for post-processing steps. Post-processing summarizes all biases and modifications that have to be done on the CG molecule and its CG coordinates. For example, an elastic network is needed to keep the tertiary structure of the protein and is applied in the post-processing stage. Finally, *martinize2* writes out the CG coordinates and the CG topology file that are production-ready.

Custom Protonation States and PTMs

Of the 20 common amino acids, there are four (GLU, ASP, LYS, HIS), which can readily change their protonation state as a function of pH or environment. Whereas commonly those amino acids are still considered to be in their pH7 protonation state, it is more appropriate to determine their local pKa from for example continuum electrostatics.[50] Subsequently the appropriate charge of the amino-acid can be determined from that pKa and set for the simulation. Even though recently more advanced methods became available for dynamically treating protonation states[51–53] - also at the Martini level[54,55] - the fixed charge approach is still the most common and for Martini most computationally efficient. However, the previous *martinize* version lacked the functionality to treat protonation states for all amino acids. Only histidine protonation states could be set interactively but only for two of three possible protonation states.

Other protonation states as defined by the atomistic structure coordinates or residue names were ignored without warning. In addition, the interactive setting of protonation states becomes very cumbersome for large protein complexes.

To overcome this problem and make protonation state handling easier and more robust, we utilize a dual strategy in *martinize2* to identify and correctly set the protonation states (see **Figure 4**). In route a) the user provides atomistic structure coordinates with AA residue names including those of non-default protonation states corresponding to the naming conventions used in CHARMM[56] or AMBER[57]. Protonation states can be obtained from online servers such as H++[58] or propKa[59], for example. If the residue names are correctly given, they can be matched against the parameters in the library and the CG residue obtains the correct protonation state. In the alternative route b), the residue name is simply that of the default pH 7 amino acid, however, the structure file contains an additional hydrogen. In the repair and identify step the chemical graph of the amino acid is compared to the building blocks in the library and any unexpected atoms are flagged. For example, in the case of protonated histidine, the additional hydrogen is labeled (see **Figure 4**). Subsequently, *martinize2* checks if there are any modifications that would match the complete input graph if added to the original block. In the Histidine example, the modification contains the additional hydrogen which together with the original histidine block make up a protonated histidine. The modification also changes the mapping such that the correct protonation state is set at the CG level. This route is more appropriate for example when processing crystal structure files, which are not necessarily named according to any force field convention. We have tested this feature on two protein structures taken from the PDB (1MJ5, 3LZT) and processed as described in the Methods section. In 1MJ5 there are six Histidine residues of which one is predicted to be charged at pH 7. The others are neutral. However, they are divided between the E-tautomer (3 residues) and the 5-tautomer (2 residues). Martini 3 parameters are different for the two tautomers in contrast to Martini2, which is accordingly recognized by *martinize2*. In addition, for lysozyme, we have considered residue GLU35 protonated, which would be appropriate at a pH of 6 or less. For both examples, the appropriate protonation states and tautomers are generated at the CG level.

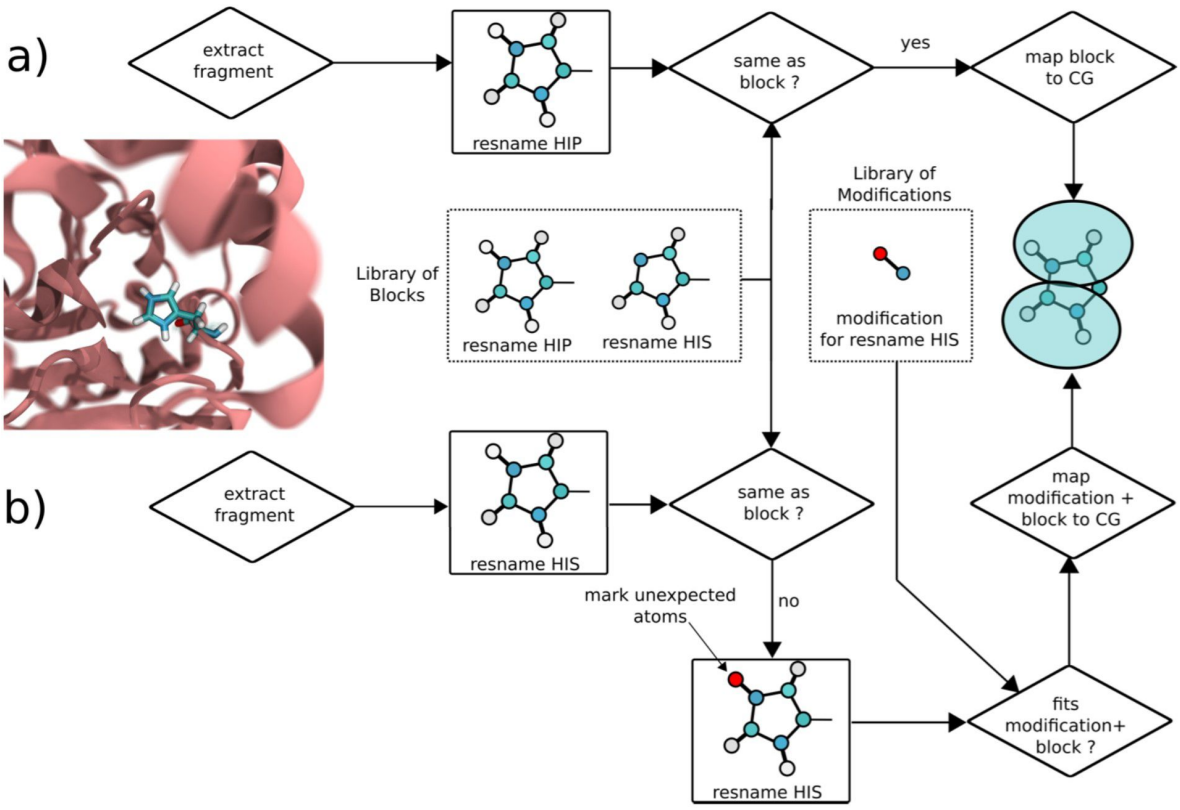


Figure 4

Workflows for identifying protonation states or PTMs exemplified on protonated histidine.

In route a) the residue name of the protonated histidine extracted from the atomistic coordinates matches the residue name in the library and matches the fragment. Hence the protonation state is correctly picked up. In route b) the residue name matches that of neutral Histidine in the library. A mismatch of the fragments is recognized and the extra hydrogen is labelled. Subsequently by matching the extra hydrogen to a modification of the histidine block the protonated Histidine is recognized as neutral Histidine plus protonation modification and the correct parameters for protonated Histidine are generated.

The same procedure used for setting protonation states also applies to identifying any other common PTM. Using this procedure, lipidation, phosphorylation, amination or acetylation can be taken into account automatically. To demonstrate that *martinize2* can handle PTMs, we have implemented dummy parameters for testing of Tyrosine phosphorylations in the M2 force field and generated a Martini topology for the EGFR kinase as an example (PDB 2GS2). Residue TYR845 (see **Figure 5**), which is located in the activation loop of the EGFR kinase, is phosphorylated when the kinase is activated.[56] *Martinize2* was able to convert the structure in one go to M2 resolution. We note that at the time of writing the M3 force field is lacking parameters for these PTMs and they are therefore not implemented in *martinize2* yet. In this case, a warning is issued by the program.

Expanding the Options of Elastic Network Fine-Tuning

Due to the limitations in most coarse-grained protein models (e.g. lack of explicit hydrogen bonding directionality), the tertiary structure has to be enforced with a structural bias called elastic network (EN).[60] An EN for Martini proteins consists of weak harmonic bonds between backbone beads of residues (within a chosen cut-off distance) and is generated after the resolution transformation as a postprocessing step.[39,41] *Martinize* offered only two types of EN options, the regular model and the Elnedyn[39] approach, both of which are also implemented in *martinize2*. However, as the EN fixes the tertiary structure, changes in the structure upon, e.g., ligand binding are not captured. To improve protein models in this aspect recently Go-like models have been applied to Martini.[40] In a Go-like model the harmonic bonds are substituted by custom Lennard-Jones (LJ) interactions that can dissociate, thereby allowing for some tertiary structure changes. Within the scope of Martini, a workflow is available to replace the elastic network with a Go model that is generated from a contact map.

Even though Go models offer better flexibility, they are currently limited to single monomeric protein units and require some fine-tuning to get the optimal performance.[40] Especially, for HT workflows the EN approach is therefore the preferred option. To further improve upon the elastic networks generated by the old *martinize*, *martinize2* offers several options to fine-tune the EN and get better behavior within the constraints of the EN approach. Besides the cut-off and forceconstant, *martinize2* now implements a residue minimum distance (RMD). The RMD is defined as a graph distance and dictates how far residues need to be apart in order to participate in elastic bonds. Defining the RMD as a graph distance means that no bonds are generated between residues that are for example bound by a disulfide bridge. It thus presents a more rigorous implementation than in the previous version. Usually, the residue minimum distance is 3 in order to avoid the EN competing with the bonds, angles, and dihedrals between the backbone beads.

We note that this is part of the Martini protein model and should not be changed. Additionally, *martinize2* allows to select which beads to generate the EN between. This option is needed for Martini 2 DNA[34], for example. Martini 2 DNA offers a stiff EN version, where also sidechain beads are included. Furthermore, *martinize2* allows to define where in the protein to apply the elastic network. This is done with the EN unit option. The EN unit can be a molecule, chain, all, or ranges of residue indices. The most trivial option is *all* in which case an EN is applied between all protein molecules in the system. The option molecule and chain yield the same network, if distinct molecules are also distinct chains. However, when two chains are connected by a disulfide bridge, for example, they would be one ***Molecule*** in the martinize sense. On the other hand, if the interface is not very well defined or more flexible, biasing the two chains separately could improve the EN. In that case, the *chain* option can be used. This use-case is shown for the human insulin dimer in **Figure 6a** and **Figure 6b**. The human insulin dimer consists of two chains, which are connected by two disulfide bridges. If the molecule or all option is used an EN is generated within the chains and between the chains (**Figure 6b**). However, to avoid generating

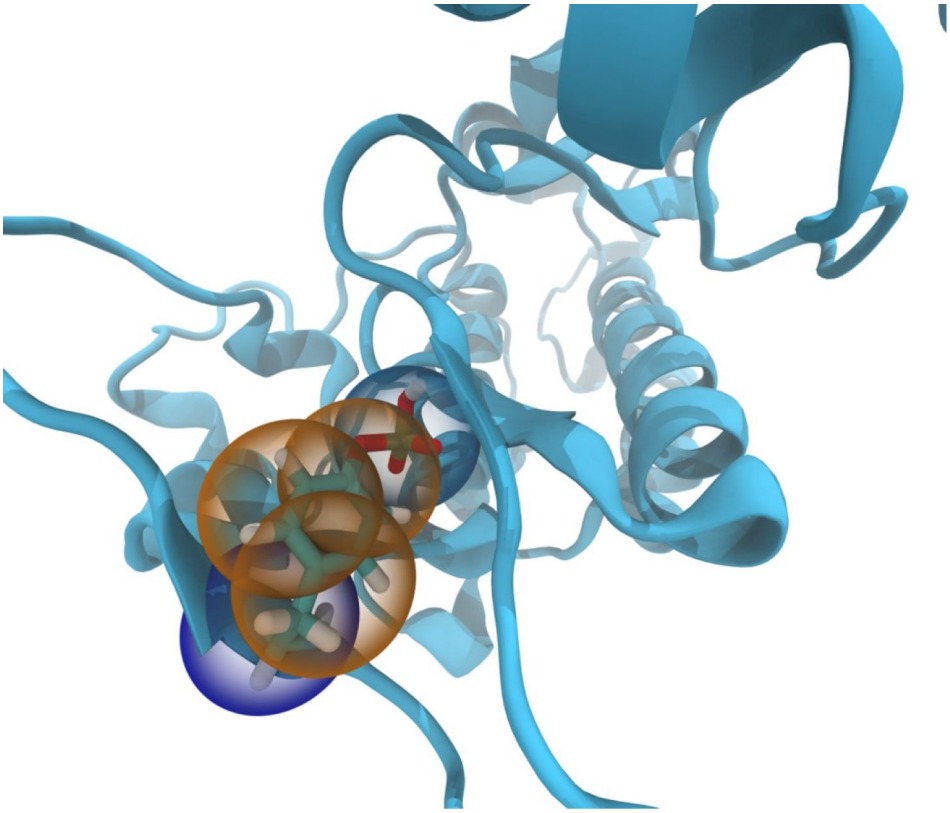


Figure 5

Example of automated identification of PTMs.

CG Martini model of phosphorylated Tyrosine found in the EGFR kinase activation loop. The mapped structure of the phosphorylated residue is shown as beads overlying the atomistic structure.

the EN between the two chains the chain option can be supplied in which case the EN is only generated within chains. As the zoom-in on the tail part shows there are no more bonds between the two chains in **Figure 6b** whereas there are in **Figure 6a**.

Furthermore, *martinize2* allows the definition of regions of residue IDs where an EN should be generated. This feature gives maximum flexibility and allows to bias structural regions of proteins whereas an EN in intrinsically disordered regions (IDRs) can be avoided. For example, **Figure 6c** and **Figure 6d** show the FtsZ protein of E-coli as predicted by alpha-fold.[61,62] FtsZ possesses a structural unit and two disordered tail domains. With the region option, *martinize2* allows the generation of an EN only for the structural domain. Within the old *martinize* superfluous bonds needed to be removed manually.

Finally, we note that *martinize2* is now implemented in the Martini Database (MAD) GUI, which offers a further utility to remove certain elastic bonds selectively.[63] We note that this option is only available for protein molecules at the moment.

Beyond Proteins: Incorporating other Molecules in *Martinize*2

Legacy *martinize* is only applicable to one category of molecule (i.e. proteins or DNA), which is one of its biggest drawbacks even for setting up simple protein simulations. *Martinize2* allows the inclusion of new classes of molecules without adjusting the codebase. For instance, proteins frequently have other molecular units associated such as ligands, cofactors, metal ions, or lipids. The general workflow of *martinize2* allows us to convert these molecules in one go provided that the library files are present. In this way, no post-hoc step is needed, that maps and parameterizes the system. Having a single step for topology generation greatly facilitates HT workflows such as protein-ligand binding, one of the cornerstones of CADD.

We test this feature on two protein complexes. The first test case concerns Flavin Reductase (see **Figure 7a**), which consists of two chains that have flavin mononucleotide ligands (FMN) and one NAD cofactor bound (2BKJ). M2 parameters and mappings from the GROMOS force field were previously published.[64] Parameters and mappings have been added to the *vermouth* database. Subsequently, the system could be converted in one step. During a short simulation, the cofactors remain well bound, indicating that no inappropriate parameters or faulty geometries were generated. Next, we created topologies and starting structures for Lysozyme with a benzene molecule bound (1L84), using the M3 force field (**Figure 7b**). The protein and ligand were again converted in one step and then simulated for a short period. As previously the ligand stays bound, showing that the protocol generates reasonable starting structures and correct parameters.

To fully leverage this new feature, ligand data files are required to be present. Thus, we implemented mappings and parameters from a previously published small molecule database for the Martini 3 force field.[65] The set comprises 43 small molecules, which are often part of drugs or drug precursors. All small molecules have corresponding parameters in the CHARMM ligand database, which allows users to directly convert atomistic CHARMM simulations to Martini. Mapping directly from crystal structures as present for example in the PDB or other force fields is also possible. In these cases, the residue names may have to be adjusted to be the same as in the CHARMM naming convention.

In addition to creating topologies for linear biopolymers, *martinize2* is now also able to handle topologically more complex molecules. For example, crown ether (**Figure 7c**) consists of six polyethylene glycol (PEG) repeat units and is cyclic. To test whether *martinize2* can handle cyclic molecules of multiple repeat units it was converted to Martini2 resolution applying the latest PEG parameters.[66] The second example is branched polyethylene (PE), where we chose a sequence that begins with two linear units followed by three branched ones and two linear units after. Also, this molecule is converted to Martini2 resolution[67] by *martinize2* (**Figure 7d**).

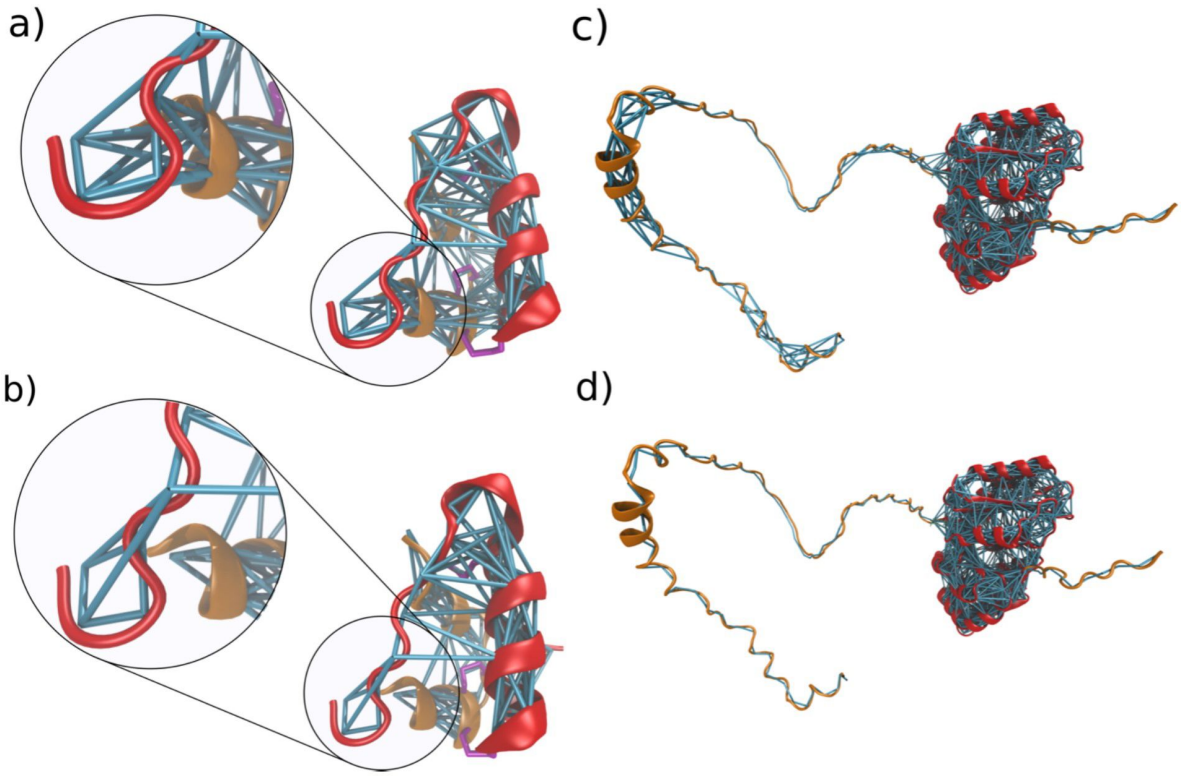


Figure 6

Fine-tuning options for the elastic network.

a) Elastic networks and backbone bonds within the human insulin dimer when generated with the molecule or all-option. The dimer consists of two chains colored in red and orange, which are connected by two disulfide bridges shown in purple. EN bonds are generated between the two chains and within the chains. b) Elastic network and backbone bonds within the insulin dimer when generated with the chain option. In this case, no elastic bonds are generated between the two chains. They are only connected by the disulfide bridge and non-bonded interactions. c) Elastic network within the Ftsz protein, when generated for both the intrinsically disordered tail domains (orange) and structural domain (red) d) Elastic network within the Ftsz protein when the EN is only generated within the structural domain by defining the EN unit as going from resid 12 to 320.

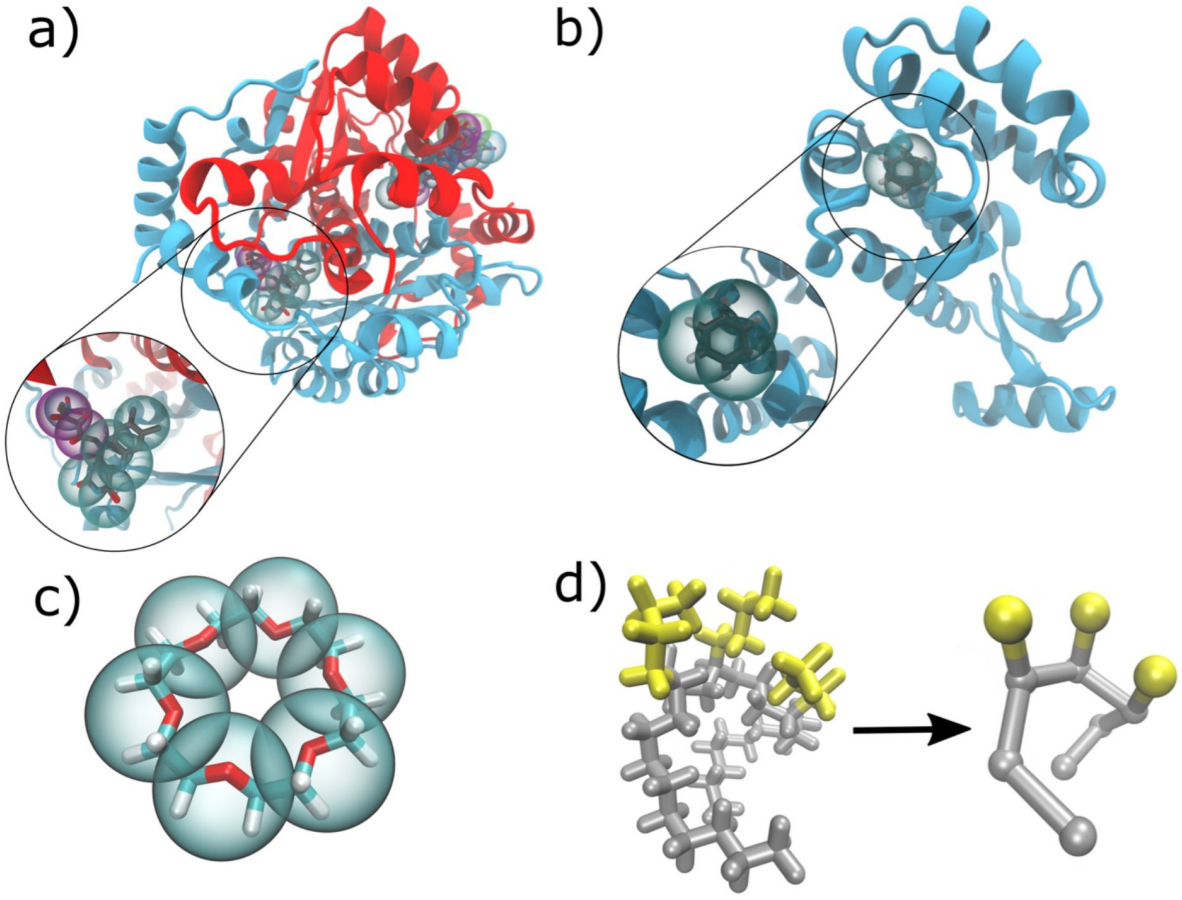


Figure 7

Ligands, cofactors, and polymers transformed to CG Martini level.

a) Flavin Reductase with two FMNs and one NDP cofactor bound in the reference all-atom state and mapped to Martini CG as indicated by the spheres. The inset shows a zoom onto FMN; b) Lysozyme with benzene ligand bound in the reference all-atom structure and mapped to Martini CG resolution; c) Crown ether with Martini beads shown on top of the all-atom structure; d) Branched polyethylene at all-atom resolution (left) and Martini resolution (right) with the linear chain part shown in gray and the branches in yellow.

Finally, we have set up instructions on how researchers can submit parameters to the database allowing it to grow and support other researchers. In addition, *martinize2* facilitates dynamic linking of citations to parameters. With this mechanism, citations are printed at the end of the run that dynamically includes citations to all parameters used in the final topology. Such a system also allows researchers to easily receive credit for contributed parameters.

Complexity Benchmark

To assess the robustness of *martinize2* in a high-throughput use case, we processed the template library used by the I-TASSER[68] protein prediction software (**Figure 8**). At the time of download (26 March 2021), the dataset contained 87084 protein structures. We processed each of these structures with *martinize2* to get M2.2 models with elastic networks. We then minimized the energy of the CG protein in a vacuum to validate that the generated structures and topology could be processed by GROMACS 2022.3.

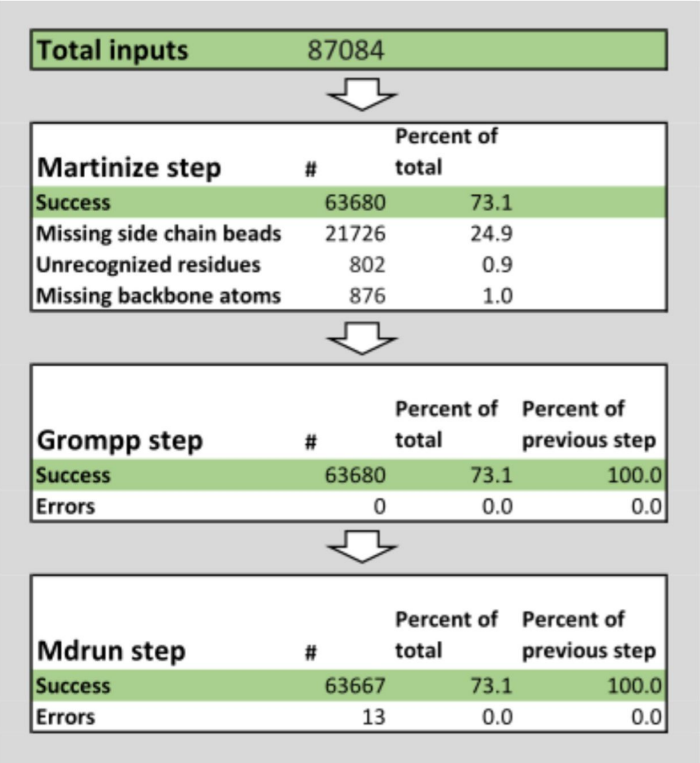
Of the 87084 structures in the dataset, 63680 (73%) could be processed through the whole workflow without error. The main cause of failure (25% of the structures) was missing coordinates in the input structures. When all the atoms that compose a bead are missing from the input, *martinize2* can generate a topology but it cannot generate coordinates for the bead. Note that if only some atoms are missing, then *vermouth* does estimate the position of the bead. 876 structures (1%) had missing coordinates in the backbone that prevented the use of DSSP[69,70]. Finally, 802 input structures (1%) had at least one residue that was inconsistent with the library. Upon further inspection, most of these structures contain malformed glycine residues with an unexpected Cp atom. *Martinize2* detected these inconsistencies and emitted a warning for each of them; warnings can be explicitly and selectively ignored, if they are not no output is written to avoid subsequent workflow steps working with corrupted files.

All the 63680 input structures that were successfully processed by *martinize*2 could be processed by the GROMACS pre-processor (*grompp*). However, 13 structures failed the energy minimization. A visual inspection of some of these failing inputs shows the input atomistic structures can be problematic. Erroneous interatomic distances (steric clashes or extended bonds) lead to high energies in the CG systems, which causes a failure in the energy minimization routine. Likely these starting structures are also not numerically stable in a subsequent simulation.

As a second test case to assess the robustness of *martinize2*, we processed a subset of the AlphaFold Protein Structure Database.[61,62] 200,000 randomly chosen unique protein structures (see Supporting Information) were given to *martinize2* and subsequently an energy minimization was performed, if the structure could successfully be converted to coarse-grained representation. Of the 200.000 structures in the dataset, 7 structures (see Supporting Information) raised an error during the conversion step. Upon further (visual) inspection of the problematic structures, we concluded that all errors were caused by inaccurate initial atomistic coordinates. These inaccurate atomic positions caused bonds to not be identified or additional superfluous bonds to be detected (**Figure 9**). In these cases, the unrecognizable residues were detected and caused *martinize2* to emit a warning. The remaining 199,993 successfully converted structures could be processed by the GROMACS pre-processor (grompp) and it was possible to perform an energy minimization.

Discussion

In the previous section, we have presented the *vermouth* python library for facilitating topology generation and manipulation. For researchers to use *vermouth* as a framework for software development it presents a clear API separated into data structures, parsers, and processors. In addition, the library relies on only three permissibly licensed open-software projects namely



Total inputs	87084

Martinize step	#	Percent of total
Success	63680	73.1
Missing side chain beads	21726	24.9
Unrecognized residues	802	0.9
Missing backbone atoms	876	1.0

Grompp step	#	Percent of total	Percent of previous step
Success	63680	73.1	100.0
Errors	0	0.0	0.0

Mdrun step	#	Percent of total	Percent of previous step
Success	63667	73.1	100.0
Errors	13	0.0	0.0

Figure 8

Summary of the successes and failures of the high-throughput pipeline.

We ran the pipeline on the 87084 structures from the template library used by the I-TASSER[68] protein prediction software of which 73% could be converted with martinize2. The other 26.4% failed mostly due to missing coordinates, and unrecognized residues. For 100% of the converted structures, a GROMACS run input file (i.e. tpr-file) could be generated, and on all but 13 of the converted structures, an energy minimization could be performed.

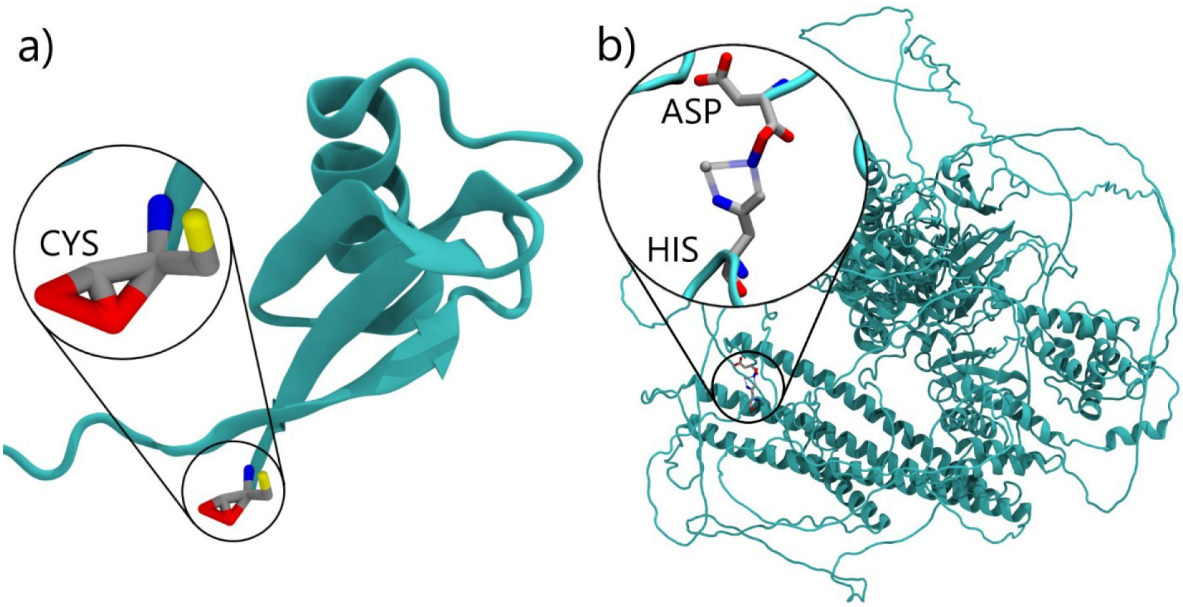


Figure 9

Two examples of problematic atomistic protein structures flagged by martinize2.

a) the cysteine residue with too small O-O and O-C distances leads to superfluous bonds being recognized. b) the incorrect interatomic distances in the histidine ring led to missing bonds (transparent), an erroneous O-N bond connecting the histidine to a neighboring asparagine. Additionally, a nitrogen atom is switched for an oxygen atom in asparagine.

numpy[71], scipy[72], and networkx[49]. This allows researchers more freedom in licensing their code and reduces the potential for bugs introduced by dependency changes. Furthermore, the central data structure represents molecules as graphs. Representing molecules as graphs allows to leverage algorithms from graph theory. Using graph theory for many of the workflows underlying the ***Processors*** makes them faster and more robust towards edge cases. Even though applying graph theory to molecules is not a new idea[73–75], *vermouth* is specifically designed to also handle coarse-grained level molecule transformation focusing on the Martini force field.

Therefore, *vermouth* presents additional functionality often lacking from other packages. For example, the handling of virtual-sites, which are ubiquitous in many M3 molecules, is rigorously handled in all ***Processors***. As another more general example, the ***Processor*** applying interactions between residues can automatically compute structural biases from the mapped coordinates. Finally, the *vermouth* library adheres to the FAIR principles[42,43] to allow adoption by non-experts and ensure quality control. In particular, for both the *vermouth* library and *martinize2*, continuous integration testing is implemented and code review is required. The software is also semantically versioned, and it is distributed through established channels, most notably the Python Package Index, and hosted openly on GitHub.

We have shown how *vermouth* was used to shape the *martinize2* program. However, *martinize2* is not the only program leveraging the power of the *vermouth* library. The *polyply* python suite is another library and collection of command line programs built upon *vermouth*. *Polyply* enables users to generate both atomistic and coarse-grained simulation input data, *i.e.* structures and topologies, from sequence information. As such, it allows building system coordinates for arbitrarily complex macromolecular systems and nanomaterials.[45] Furthermore, the *martini-sour* package[54,76], which is currently under development, utilizes *vermouth* to convert topology files from regular Martini to titratable Martini simulations. These examples already illustrate that *vermouth* has the potential to indeed become the central framework for Martini software development and possibly for other scientific software developments.

Martinize2 enables researchers to prepare simulation input files for arbitrary (bio)polymers, starting from atomistic structure. We have shown in-depth examples focusing on protein-specific applications, given that they are the most important target for *martinize2*. However, also more complex non-biological molecules such as a cyclic crown ether and branched PE were showcased to demonstrate the capabilities of *martinize2*. Furthermore, the user has complete control over the data files used. The abstraction of force field data into Blocks, Modifications, and Links allows researchers to reason about model intricacies in a structured manner. This helps the development of optimized models and parameters for complex (polymeric) molecules, as well as clearly defining in which combinations these are validated. The new program uses algorithms from graph theory to identify atoms and assign the appropriate interactions. This makes the program more tolerant towards its input so that the users have to worry less over details such as atom names, or ensuring that all residues are in order and appropriately numbered. In particular, *martinize2* is capable of detecting and using protonation states and PTMs and capping groups automatically. In addition, *martinize2* allows to fine-tune the EN and as it is not limited to proteins can also generate parameters for ligands, cofactors, or lipids.

In practice, there are decisions a user needs to make when using *vermouth* and *martinize2*, especially for HT pipelines. *Martinize2* detects but does not reconstruct atoms that are missing from the input structures; these missing atoms can have adverse effects on the result. In the most harmless cases, they only shift the position of a particle in the output structure. When all the atoms for a particle are missing, then the program cannot compute a position for that particle leading to an incomplete output where a particle does not have coordinates. Also, some workflows depend on DSSP[69,70] to assign secondary structures and some specific missing atoms can prevent DSSP from working properly. In those cases, *martinize2* issues a warning whenever it cannot automatically take care of pitfalls. Handling of these cases is a central difference between

the new and old version. The old version either terminates with an undefined error or, probably worse, runs and gives output that does not correspond to the atomistic structure given as input. To illustrate the robustness of *martinize2* towards problematic input, we applied the program to the complete I-TASSER database (~87k structures) as well as a subset of the AlphaFold Protein Structure Database (~200k structures). For the two benchmark cases, *martinize2* was able to issue a warning or error for all structures, which contained seriously malformed residues. Of the first database only 13 structures failed in the energy minimization due to problematic starting coordinates but not obviously malformed residues. In the second benchmark set only 7 seriously malformed residues were identified, and all other structures were successfully energy minimized. Thus, we consider *martinize2* more robust and fit for HT and high-complexity tasks.

Ultimately, the robustness comes at a price. *Martinize2* uses a subgraph isomorphism to identify atoms based on their connectivity, and then issues a warning or repairs the input. However, subgraph isomorphism is an NP-complete problem[77]. As a result, *martinize2* is significantly slower than *martinize*. Nevertheless, considering the flexibility the new program offers, in addition to the fact that it is still fast enough to process all entries in the I-TASSER data bank [68], this is deemed to be acceptable. Even though *martinize2* will most likely never be as fast as *martinize* we note that many of the processes can still be optimized to yield further performance increases. Aside from the performance limitations, *vermouth,* and *martinize2* present some other limitations as well. Both *martinize2* and *vermouth* are currently only capable of writing topologies in GROMACS format. However, our library does not use the MD parameters of the produced topologies or calls GROMACS functions, so support for other MD engines can be added in the future. In addition, since *vermouth* defines an API, it could even be integrated with existing software such as OpenMM.[78] Furthermore, the processor pipeline underlying *martinize2* is currently hardcoded. Future improvements will focus on making the workflow defined by *martinize2* more flexible, in order to include the processor pipeline as part of the force field definitions. This would enable the use of different pipelines for different force fields, allowing for easier force field-specific post-processing.

Methods

Preparation of protein input files

Crystal structures were obtained from the RCSB for the following proteins (3LZT; 2GS2; 2BKJ; 1L84; 3I40; 3IGM, 1MJ5) or the Alpha Fold Data Bank[61] for FtsZ with the ID A0A7Y6D765. Hydrogens and missing heavy atoms were reconstructed using the PRAS package, if appropriate.[79] For 3LST and 1MJ5, the pKa and half-way titration point were estimated using the propka package.[59] For 3LST the GLU35 was protonated using the CHARMM-GUI solution builder.[35],[80] The HIS-tag of 1MJ5 was removed.

AA simulations

For 2GS2 and 1L84 CHARMM AA parameters[56] were created using the CHARMM-GUI solution builder[35],[80] and a small equilibration simulation (20ns) was run before the structures were converted with *martinize2*. The AA simulation used the recommended nonbonded force settings as for CHARMM with GROMACS. [81] The temperature was maintained using the v-rescale thermostat by Bussi *et al.*[82] at 310K and pressure was maintained at 1 bar using the Parrinello-Rahman[83] barostat (t = 12 ps) after initial equilibration with the Berendsen[84] barostat.

CG simulations

All CG MD simulations were run using GROMACS 2021.5[85] and the recommended mdp parameters for Martini 2[86] and Martini 3[13] respectively. In particular, the Lennard-Jones interactions were cut-off at 1.1 nm and electrostatics were treated with reactionfield (cut-off 1.1

nm, dielectric constant 15). The time-step was 20 fs in all cases and the production trajectories were run with the standard leap-frog integrator. Temperature was maintained using the v-rescale thermostat by Bussi *et al.*[82] at 310K with (t = 6 ps) and separate coupling groups for solvent and proteins. The pressure was maintained at 1 bar using the Berendsen barostat for equilibrations (t=6ps). The initial systems were solvated using the polyply[45] package or gmx solvate utility.

Complexity benchmark

The (Swiss-Prot) subset of the AlphaFold protein structure database used for the complexity benchmark contained 542.378 pdb structure files at the time of download (22 December 2022). The testing pipeline we used was written in Python and randomly picked 200.000 structures which were given to *martinize2*. Possible errors during conversion or the subsequent grompp and energy minimization steps were captured.

Code availability

All code can be found online at *https://www.github.com/marrink-lab/vermouth-martinize*. In addition, all released versions are also published on the Python Package Index at *https://www.pypi.org/project/vermouth*. The documentation is available at *https://vermouth-martinize.readthedocs.io/en/latest/index.html*.

Data availability

Input files and commands required to reproduce the example test cases from this paper are available on GitHub at *https://github.com/marrink-lab/martinize-examples*. MD trajectories and benchmark data are available upon reasonable request from the corresponding authors.

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Author contributions

PCK and SJM conceived the project; PCK, JB, and FG implemented the described software; PCK, JB, TAW designed the program structure; PCTS & FG designed the benchmark tests used along the development of the code to guarantee the accuracy of the models; MvT ran the protein benchmark test-case using the alpha-fold data-base, while JB ran and analyzed the iTasser data-base; FG ran

and analyzed all other test-cases. PCTS helped to implement the force field files, and managed feedback from beta testers; PCK and FG wrote the manuscript, with contributions from all authors. SJM provided guidance and supervision in the project.

Competing interests

The authors declare no competing interests.

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Editors

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Reviewer #1 (Public Review):

Summary:

In this study, the authors provide a new computational platform called Vermouth to automate topology generation, a crucial step that any biomolecular simulation starts with. Given a wide arrange of chemical structures that need to be simulated, varying qualities of structural models as inputs obtained from various sources, and diverse force fields and molecular dynamics engines employed for simulations, automation of this fundamental step is challenging, especially for complex systems and in case that there is a need to conduct high-throughput simulations in the application of computer-aided drug design (CADD). To overcome this challenge, the authors develop a programing library composed of components

that carry out various types of fundamental functionalities that are commonly encountered in topological generation. These components are intended to be general for any type of molecules and not to depend on any specific force field and MD engines. To demonstrate the applicability of this library, the authors employ those components to re-assemble a pipeline called Martinize2 used in topology generation for simulations with a widely used coarse-grained model (CG) MARTINI. This pipeline can fully recapitulate the functionality of its original version Martinize but exhibit greatly enhanced generality, as confirmed by the ability of the pipeline to faithfully generate topologies for two high-complexity benchmarking sets of proteins.

Strengths:

The main strength of this work is the use of concepts and algorithms associated with induced subgraph in graph theory to automate several key but non-trivial steps of topology generation such as the identification of monomer residue units (MRU), the repair of input structures with missing atoms, the mapping of topologies between different resolutions, and the generation of parameters needed for describing interactions between MRUs. In addition, the documentation website provided by the authors is very informative, allowing users to get quickly started with Vermouth.

Weaknesses:

Although the Vermouth library is designed as a general tool for topology generation for molecular simulations, only its applications with MARTINI have been demonstrated in the current study. Thus, the claimed generality of Vermouth remains to be exmained. The authors may consider to point out this in their manuscript.

https://doi.org/10.7554/eLife.90627.2.sa2

Reviewer #2 (Public Review):

This work introduces a Vermouth library framework to enhance software development within the Martini community. Specifically, it presents a Vermouth-powered program, Martinize2, for generating coarse-grained structures and topologies from atomistic structures. In addition to introducing the Vermouth library and the Martinize2 program, this paper illustrates how Martinize2 identifies atoms, maps them to the Martini model, generates topology files, and identifies protonation states or post-translational modifications. Compared with the prior version, the authors provide a new figure to show that Martinize2 can be applied to various molecules, such as proteins, cofactors, and lipids. To demonstrate the general application, Martinize2 was used for converting 73% of 87,084 protein structures from the template library, with failed cases primarily blamed on missing coordinates.

I was hoping to see some fundamental changes in the resubmitted version. To my disappointment, the manuscript remains largely unchanged (even the typo I pointed out previously was not fixed). I do not doubt that Martinize2 and Vermouth are useful to the Martini community, and this paper will have some impact. The manuscript is very technical and limited to the Martini community. The scientific insight for the general coarse-grained modeling community is unclear. The goal of the work is ambitious (such as high-throughput simulations and whole-cell modeling), but the results show just a validation of Martinize2. This version does not reverse my previous impression that it is incremental. As I pointed out in my previous review (and no response from the authors), all the issues associated with the Martini model are still there, e.g. the need for ENM. In this shape, I feel this manuscript is suitable for a specialized journal in computational biophysics or stays as part of the GitHub repository.

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Reviewer #3 (Public Review):

The manuscript Kroon et al. described two algorithms, which when combined achieve high throughput automation of "martinizing" protein structures with selected protonation states and post-translational modifications. After the revisions provided by the authors, I recommend minor revision.

The authors have addressed most of my concerns provided previously. Specifically, showcasing the capability of coarse-graining other types of molecules (Figure 7) is a useful addition, especially for the booming field of therapeutic macrocycles.

My only additional concern is that to justify Martinize2 and Vermouth as a "high-throughput" method, the speed of these tools needs to be addressed in some form in the manuscript as a guideline to users.

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Author response:

The following is the authors' response to the original reviews.

> ***Reviewer #1 (Public Review):***
>
> *Summary:*
>
> *In this study, the authors provide a new computational platform called Vermouth to automate topology generation, a crucial step that any biomolecular simulation starts with. Given a wide arrange of chemical structures that need to be simulated, varying qualities of structural models as inputs obtained from various sources, and diverse force fields and molecular dynamics engines employed for simulations, automation of this fundamental step is challenging, especially for complex systems and in case that there is a need to conduct high-throughput simulations in the application of computer-aided drug design (CADD). To overcome this challenge, the authors develop a programming library composed of components that carry out various types of fundamental functionalities that are commonly encountered in topological generation. These components are intended to be general for any type of molecules and not to depend on any specific force field and MD engines. To demonstrate the applicability of this library, the authors employ those components to re-assemble a pipeline called Martinize2 used in topology generation for simulations with a widely used coarse-grained model (CG) MARTINI. This pipeline can fully recapitulate the functionality of its original version Martinize but exhibit greatly enhanced generality, as confirmed by the ability of the pipeline to faithfully generate topologies for two high-complexity benchmarking sets of proteins.*
>
> *Strengths:*
>
> *The main strength of this work is the use of concepts and algorithms associated with induced subgraph in graph theory to automate several key but non-trivial steps of topology generation such as the identification of monomer residue units (MRU), the repair of input structures with missing atoms, the mapping of topologies between different resolutions, and the generation of parameters needed for describing interactions between MRUs.*

> *Weaknesses:*
>
> *Although the Vermouth library appears promising as a general tool for topology generation, there is insufficient information in the current manuscript and a lack of documentation that may allow users to easily apply this library. More detailed explanation of various classes such as Processor, Molecule, Mapping, ForceField etc. that are mentioned is still needed, including inputs, output and associated operations of these classes. Some simple demonstration of application of these classes would be of great help to users. The formats of internal databases used to describe reference structures and force fields may also need to be clarified. This is particularly important when the Vermouth needs to be adapted for other AA/CG force fields and other MD engines.*

We thank the reviewer for pointing out the strengths of the presented work and agree that one of the current limitations is the lack of documentation about the library. In the revision, we point more clearly to the documentation page of the Vermouth library, which contains more detailed information on the various processors. The format of the internal databases has also been added to the documentation page. Providing a simple demonstration of applications of these classes is a great suggestion, however, we believe that it is more convenient to provide those in the form of code examples in the documentation or for instance jupyter notebooks rather than in the paper itself.

> *The successful automation of the Vermouth relies on the reference structures that need to be pre-determined. In case of the study of 43 small ligands, the reference structures and corresponding mapping to MARTINIcompatible representations for all these ligands have been already defined in the M3 force field and added into the Vermouth library. However, the authors need to comment on the scenario where significantly more ligands need to be considered and other force fields need to be used as CG representations with a lack of reference structures and mapping schemes.*

We acknowledge that vermouth/martinize2 is not capable of automatically generating Martini mappings or parameters on the fly for unknown structures that are not part of the database. However, this capability is not the purpose of the program, which is rather to distribute and manage existing parameters. Unlike atomistic force fields, which frequently have automated topology builders, Martini parameters are usually obtained for a set of specific molecules at a time and benchmarked accordingly. As more parameters are obtained by researchers, they can be added to the vermouth library via the GitHub interface in a controlled manner. This process allows the database to grow and in our opinion will quickly grow beyond the currently implemented parameters. Furthermore, the API of Vermouth is set up in a way that it can easily interface with automated topology builders which are currently being developed. Hence this limitation in our view does not diminish the applicability of vermouth to high-throughput applications with many ligands. The framework is existing and works, now only more parameters have to be added.

> ***Reviewer #2 (Public Review):***
>
> *Summary:*
>
> *This manuscript by Kroon, Grunewald, Marrink and coworkers present the development of Vermouth library for coarse grain assignment and parameterization and an updated version of python script, the Martinize2 program, to build Martini coarse grained (CG) models, primarily for protein systems.*
>
> *Strengths:*

> *In contrast to many mature and widely used tools to build all-atom (AA) models, there are few well-accepted programs for CG model constructions and parameterization. The research reported in this manuscript is among the ongoing efforts to build such tools for Martini CG modeling, with a clear goal of high-throughput simulations of complex biomolecular systems and, ultimately, whole-cell simulations. Thus, this manuscript targets a practical problem in computational biophysics. The authors see such an effort to unify operations like CG mapping, parameterization, etc. as a vital step from the software engineering perspective.*
>
> *Weaknesses:*
>
> *However, the manuscript in this shape is unclear in the scientific novelty and appears incremental upon existing methods and tools. The only "validation" (more like an example application) is to create Martini models with two protein structure sets (I-TASSER and AlphaFold). The success rate in building the models was only 73%, while the significant failure is due to incomplete AA coordinates. This suggests a dependence on the input AA models, which makes the results less attractive for high-throughput applications (for example, preparation/creation of the AA models can become the bottleneck). There seems to be an improvement in considering the protonation state and chemical modification, but convincing validation is still needed. Besides, limitations in the existing Martini models remain (like the restricted dynamics due to the elastic network, the electrostatic interactions or polarizability).*

We thank the reviewer for pointing out the strengths of the presented work, but respectfully disagree with the criticism that the presented work is only incremental upon existing methods and tools. All MD simulations of structured proteins regardless of the force field or resolution rely on a decent initial structure to produce valid results. Therefore, failure upon detection of malformed protein input structures is an *essential* feature for any high-throughput pipeline working with proteins, especially considering the computational cost of MD simulations. We note that programs such as the first version of Martinize generate reasonable-looking input parameters that lead to unphysical simulations and wasted CPU hours.

The alpha-fold database for which we surveyed 200,000 structures only contained 7 problematic structures, which means that the success rate was 99% for this database. This example simply shows that users potentially have to add the step of fixing atomistic protein input structures, if they seek to run a high-throughput pipeline.

But at least they can be assured that martinize2 will make sure to check that no issues persist.

Furthermore, we note that the manuscript does not aim to validate or improve the existing Martini (protein) models. All example cases presented in the paper are subject to the limitations of the protein models for the reason that martinize2 is only the program to generate those parameters. Future improvements in the protein model, which are currently underway, will immediately be available through the program to the broader community.

> ***Reviewer #3 (Public Review):***
>
> *Summary:*
>
> *The manuscript Kroon et al. described two algorithms, which when combined achieve high throughput automation of "martinizing" protein structures with selected protonation states and post-translational modifications.*
>
> *Strengths:*

> *A large scale protein simulation was attempted, showing strong evidence that authors' algorithms work smoothly.*
>
> *The authors described the algorithms in detail and shared the open-source code under Apache 2.0 license on GitHub. This allows both reproducibility of extended usefulness within the field. These algorithms are potentially impactful if the authors can address some of the issues listed below.*

We thank the reviewer for pointing out the strengths.

> *Weaknesses:*
>
> *One major caveat of the manuscript is that the authors claim their algorithms aim to "process any type of molecule or polymer, be it linear, cyclic, branched, or dendrimeric, and mixtures thereof" and "enable researchers to prepare simulation input files for arbitrary (bio)polymers". However, the examples provided by the manuscript only support one type of biopolymer, i.e. proteins. Despite the authors' recommendation of using polyply along with martinize2/vermouth, no concrete evidence has been provided to support the authors' claim. Therefore, the manuscript must be modified to either remove these claims or include new evidence.*

We acknowledge that the current manuscript is largely protein-centric. To some extent this results from the legacy of *martinize* version 1, which was also only used for proteins. However, to show that martinize2 also works for cyclic as well as branched molecules we implemented two additional test cases and updated formerly Figure 6 and now Figure 7. Crown ether is used as an example of a cyclic molecule whereas a small branched polyethylene molecule is a test case for branching. Needless to say both molecules are neither proteins nor biomolecules.

> *Method descriptions on Martinize2 and graph algorithms in SI should be core content of the manuscript. I argue that Figure S1 and Figure S2 are more important than Figure 3 (protonation state). I recommend the authors can make a workflow chart combining Figure S1 and S2 to explain Martinize2 and graph algorithms in main text.*

The reviewer's critique is fair. Given the already rather large manuscript, we tried to strike a balance between describing benchmark test cases, some practical usage information (e.g. the Histidine modification), and the algorithmic library side of the program. In particular, we chose to add the figure on protonation state, because how to deal with protonation states—in particular, Histidines—was amongst the top three raised issues by users on our GitHub page. Due to this large community interest, we consider the figure equally important. However, we moved Figure S1 from the Supporting Information into the manuscript and annotated the already mentioned text with the corresponding panels to more clearly illustrate the underlying procedure.

> *In Figure 3 (protonation state), the figure itself and the captions are ambiguous about whether at the end the residue is simply renamed from HIS to HIP, or if hydrogen is removed from HIP to recover HIS.*

Using either of the two routes yields the same parameters in the end, which are for the protonated Histidine. In the second route, the extra hydrogen on Histidine is detected as an additional atom and therefore a different logic flow is triggered. Atoms are never removed, but only compounded to a base block plus modification atoms. We adjusted the figure caption to point this out more clearly.

> *In "Incorporating a Ligand small-molecule Database", the authors are calling for a community effort to build a small-molecule database. Some guidance on when the current database/algorithm combination does or does not work will help the community in contributing.*

Any small molecule not part of the database will not work. However, martinize2 will quickly identify if there are missing components of the system and alert the users. At that point, the users can decide to make their files, guided by the new documentation pages.

> *A speed comparison is needed to compare Martinize2 and Martinize.*

We respectfully disagree that a speed comparison is needed. We already alerted in the manuscript discussion that martinize2 is slower, since it does more checks, is more general, and does not only implement a single protein model.

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